

Detecting Fraudulent Transactions

Data Engineering with ETL (SSIS) & SSMS

Problem Statement & Objectives

2026-07-01

Background

- Financial institutions process thousands of transactions daily.
- A small number are fraudulent — large amounts, suspended accounts, or country mismatches.
- Manual review of every transaction is impossible.
- We need an automated ETL pipeline to extract, apply fraud rules, and flag suspicious records.

Problem Statement

Given a set of customer and transaction records, build an ETL pipeline that identifies and flags potentially fraudulent transactions based on defined business rules, and loads the flagged records into a staging table for further investigation.

Objectives — What We Need to Achieve

- Extract — Load Customers.csv and Transactions.csv into SQL Server staging tables using SSIS.
- Transform — Join transactions to customers and apply fraud-detection rules:
 - Transactions on suspended accounts.
 - Amount above a threshold (e.g. > \$5,000).
 - Location does not match the customer's country.
- Load — Write all flagged transactions into a FlaggedTransactions table.
- Analyse — Run SQL queries in SSMS to review and summarise flagged records.

Scope & Data Sources

Scope

- In scope: Rule-based detection using ETL (SSIS) and SQL (SSMS).
- Out of scope: Machine-learning models, real-time streaming, production deployment.

Data Sources

File	Description
Customers.csv	Customer master data (ID, name, country, account status)
Transactions.csv	Transaction records (ID, customer, amount, date, location)

Fraud Detection Rules

Rule ID	Condition	Reason
R1	AccountStatus = 'Suspended'	Activity on a blocked account
R2	Amount > 5000	Unusually high transaction value
R3	Location <> Customer.Country	Geographic mismatch

Expected Outcome & Success Criteria

Expected Outcome

- A working ETL pipeline producing a FlaggedTransactions table of every rule-violating transaction, ready for analyst review.

Success Criteria

- All CSV data loads into SQL Server without errors.
- Known planted anomalies (T1004, T1006, T1008) are correctly flagged.
- Flagged results are queryable and summarised in SSMS.

Tools & Technologies

SSMS

Database, tables, and queries

SSIS

ETL pipeline (extract, transform, load)

CSV Files

Source data